An Autoencoder and Generative Adversarial Networks Approach for Multi-Omics Data Imbalanced Class Handling and Classification

ABSTRACT

In enhancing medical diagnostics, integrating advanced machine learning has shown promise. Molecular biology has seen a surge in data from multi-omics sequencing, presenting challenges for traditional statistical methods due to high dimensionality. Most of this data is redundant, which can be reduced without significant information loss using dimensionality reduction techniques. Another challenge is class imbalance in medical datasets, leading to biased machine learning results. This study addresses these issues using a neural network with autoencoders to extract latent features and GANs to generate synthetic samples. The model begins with feature selection and predicts cancer outcomes, achieving 95.09% accuracy for bladder cancer and 88.82% for breast cancer.

MATERIALS & METHODS

As a first step, we pre-process the three omics dataset using the autoencoder to extract latent space, and then generate more minor class samples to address the imbalanced classes. The proposed model can be seen schematically in Figure 1.

Materials

The proposed model is applied to two publicly available datasets, the first is the Cancer Genome Atlas (TCGA) Breast Invasive Carcinoma (BRCA) dataset, which contains DNA, CNA, and Gene information. The second is the TCGA Bladder Urothelial Carcinoma (BLCA) dataset, which contains DNA, CNA and mRNA. For the BRCA dataset, we compared the result of applying the proposed model with the state of art SMOTE-SVM-RBF model. While for the BLCA dataset, we compared the results with NMF-GA model.

Methods

1- Autoencoder

The encoder, implemented as a dense neural network layer with rectified linear unit (ReLU) activation, compresses the input data into a lower-dimensional representation, this layer reduces the dimensionality of the input while capturing its latent space.

The mathematical representation of the encoder is shown in equation:

$$Encoder: h = f_{encoder}(x) = \phi(W_1 x + b_1)$$

Where:

- x is the input data,
- h is the encoded representation (latent space),
- $f_{encoder}$ is the encoder function,
- W_1 and b_1 are the weights and biases of the encoder layer, respectively,
- ϕ is a ReLU activation function, which returns the maximum between and 0.

The decoder, a dense neural network layer with Sigmoid activation, reconstructs the original input from the encoded representation. The mathematical representation of the decoder is show in equation :

$$Decoder: x' = f_{decoder}(h) = \phi(W_2 h + b_2)$$

Where:

- x' is the reconstructed output,
- $f_{decoder}$ is the decoder function,
- W_2 and b_2 are the weights and biases of the decoder layer, respectively.

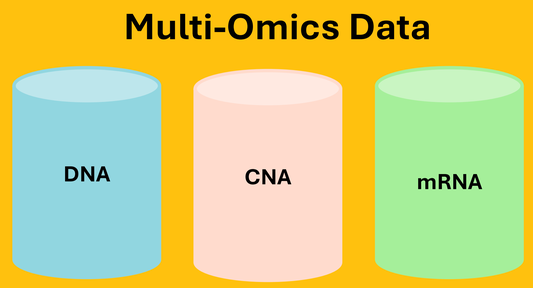
Can We Revolutionize Cancer Diagnostics with Machine Learning?

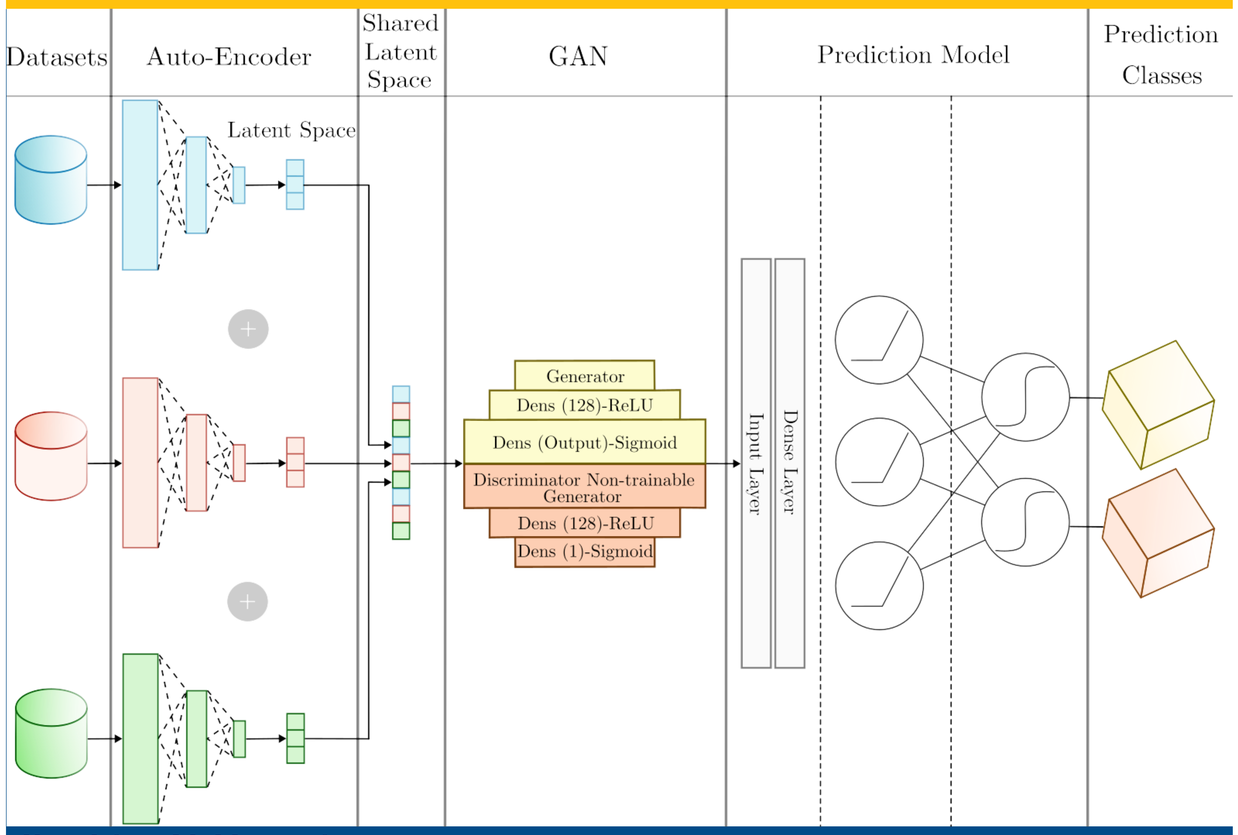
Bladder **Breast**

Our Novel Approach Achieves 95% Accuracy!

The proposed model consist of:

- ❑ Multi-Omics Data
- ❑ Autoencoder
  - ✓ Encoder
  - ✓ Decoder
- ❑ Generative Adversarial Networks
  - ✓ Generator
  - ✓ Discriminator





2- Generative Adversarial Networks (GAN)

GAN is utilized to up-sample the minor class from the shared latent space samples.

The generator network is defined as:

$$G(z) = \sigma(\phi(W_g z + b_g))$$

For the discriminator network:

$$D(x) = \sigma(\phi(W_d x + b_d))$$

The objective of the GAN is to train the generator to produce samples that are indistinguishable from real samples, and to train the discriminator to distinguish between real and fake samples.

3- Prediction Model

Firstly, we split the data into training and testing sets based on a split ratio of 0.8, that is used to construct the prediction model. A network is built of two dense layers. The first is with 128 neurons and ReLU activation function, and the other with 1 neuron and Sigmoid activation function. The model utilizes Adam optimizer[15] and binary cross-entropy loss function to optimize the accuracy. The model is trained with parameters specifying the number of training epochs of 10, batch size of 32, and a validation split of 0.2. The Sigmoid function returns the probability of the class which represents the predicted class.

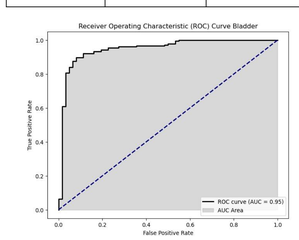
RESULTS

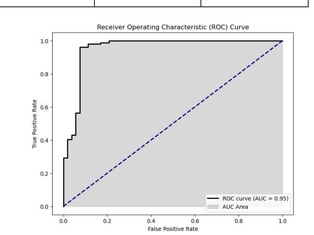
Table 1 summarizes The performance measurements for the BRCA AND BLCA dataset

Classifier	Average Accuracy	Average AUC (µm)	Precision	Recall	Average F1-score
Proposed Model	0.95098	1.0000	1.0000	0.81481	0.89796
SMOT-SVM-RBF	0.78036	0.78050	0.7758	0.8024	0.78840

Table 2 summarizes The performance measurements for the BLCA dataset.

Classifier	Average accuracy	Average AUC (µm)	Precision	Recall	Average F1-score
Proposed Model	0.88827	0.91211	0.85417	0.75926	0.80392
NMF-GA	0.78036	0.78050	0.7758	0.8024	0.78840





CONCLUSION

This study demonstrates significant advancements in predictive accuracy for cancer outcome prediction. The model augment the dataset by generating synthetic samples for the minorit class, thus mitigating class imbalance. In the final step, th approach classifies multi-omics data better for cance prediction.

Overall, our study shows how autoencoders and GANs ca enhance machine learning models' predictive capabilities fo medical diagnosis. In addition to addressing the challenge posed by imbalanced class distributions and high dimensiona omics data, our approach contributes to precision medicine an personalized healthcare.

Ibrahim Al-Hurani,
Dr. Abedalrhman Alkhateeb,
Dr. Salama Ikki

https://arxiv.org/abs/2405.09756